

SPRAGUE, MB, Canada

WMO: 714490

Lat: 49.0236N

Lon: 95.5983W

Elev: 329

StdP: 97.44

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-30.3	-37.3	0.1	-33.4	-33.8	0.2	-30.1	10.3	-13.6	9.2	-14.3	1.8	230	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	29.3	21.3	27.6	20.0	26.2	19.1	22.9	27.3	21.6	25.9	20.4	24.5	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.7	25.8	20.1	15.4	24.1	18.9	14.3	22.9	69.6	27.3	64.6	25.9	60.4	24.5	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.1	7.1	-38.0	32.3	4.0	1.3	-40.9	33.3	-43.2	34.1	-45.4	34.8	-48.3	35.8		
(5)			-38.4	24.7	4.0	1.4	-41.2	25.7	-43.6	26.6	-45.8	27.3	-48.7	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-15.6	-13.3	-5.1	3.5	10.5	16.3	18.9	17.7	12.9	5.5	-3.4	-11.8		
	DBStd	13.46	8.56	7.91	7.75	5.68	4.75	3.66	3.09	3.34	4.64	4.91	6.64	7.76		
	HDD10.0	3430	794	651	469	204	51	2	0	0	22	156	403	676		
	HDD18.3	5676	1053	884	727	446	246	81	31	52	171	398	653	934		
	CDD10.0	909	0	0	1	9	68	191	276	238	109	17	1	0		
	CDD18.3	111	0	0	0	0	3	19	48	32	8	1	0	0		
	CDH23.3	1201	0	0	0	3	57	213	485	337	101	6	0	0		
	CDH26.7	253	0	0	0	0	9	41	110	72	20	1	0	0		
(14) Wind	WSAvg	3.1	3.1	3.2	3.3	3.6	3.5	2.8	2.6	2.4	2.8	3.2	3.2	3.0		
(15) Precipitation	PrecAvg	622	23	19	29	40	67	100	96	84	66	42	28	23		
	PrecMax	798	83	56	75	106	152	210	195	255	140	130	98	61		
	PrecMin	361	3	3	3	0	12	5	37	16	8	8	0	5		
	PrecStd	112	17	14	18	28	37	54	37	60	43	28	23	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	5.7	16.5	22.8	28.0	30.4	31.5	31.3	29.3	23.4	14.7	4.2		
		MCWB	2.0	3.0	10.4	12.5	17.4	22.4	22.6	22.5	20.7	15.2	9.5	1.9		
	2%	DB	0.8	3.0	10.2	19.0	24.8	27.6	29.4	28.6	26.1	19.7	10.2	1.9		
		MCWB	-0.3	1.2	6.3	10.3	15.1	19.6	21.6	20.7	18.7	13.6	6.6	0.6		
	5%	DB	-1.2	0.9	7.2	16.1	22.2	25.7	27.7	26.8	23.8	16.5	7.7	0.4		
		MCWB	-2.2	-0.3	4.1	8.8	13.8	18.3	20.6	19.9	17.2	11.5	5.0	-0.5		
	10%	DB	-3.6	-1.2	4.5	13.4	19.7	24.0	26.1	25.1	21.3	13.5	5.1	-1.4		
		MCWB	-4.3	-2.3	2.2	7.2	12.6	17.6	19.7	18.7	16.2	9.7	3.2	-2.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.0	3.5	10.9	13.2	19.2	24.1	24.6	24.1	22.2	16.8	10.1	2.2		
		MCDB	3.9	4.9	14.3	20.7	24.5	28.5	29.0	28.7	26.9	20.0	13.0	3.7		
	2%	WB	-0.2	1.4	6.7	11.2	17.0	21.6	23.1	22.2	20.1	14.2	7.4	0.7		
		MCDB	0.7	2.7	9.6	17.6	22.0	25.8	27.3	26.8	23.9	19.0	9.7	1.6		
	5%	WB	-2.1	-0.1	4.2	9.4	15.2	19.6	21.8	21.0	18.2	11.9	5.1	-0.5		
		MCDB	-1.2	0.9	6.9	14.8	19.5	23.5	26.0	25.2	21.8	15.5	7.4	0.4		
	10%	WB	-4.3	-2.2	2.3	7.7	13.7	18.3	20.6	19.9	16.7	9.8	3.1	-2.1		
		MCDB	-3.6	-1.3	4.3	13.2	18.5	22.3	24.8	23.6	20.6	13.1	5.0	-1.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	11.8	11.3	11.7	12.2	11.4	12.1	12.5	12.0	9.7	8.4	9.0		
		MCDBR	10.0	11.1	12.1	16.7	16.2	14.3	14.0	14.6	15.5	14.8	11.6	9.4		
		MCWBR	9.0	9.3	8.4	9.8	8.4	7.7	7.6	7.9	8.8	9.0	8.1	8.1		
	5% WB	MCDBR	9.6	10.2	11.7	15.0	12.7	11.7	12.3	12.7	12.9	13.0	10.4	8.9		
		MCWBR	8.7	8.6	8.3	9.3	7.7	7.3	7.6	7.8	8.1	8.6	7.6	7.8		
(40) Clear-Sky Solar Irradiance	taub	0.262	0.295	0.293	0.343	0.359	0.376	0.380	0.374	0.346	0.303	0.275	0.256	0.256		
	taud	2.289	2.193	2.404	2.381	2.400	2.379	2.386	2.411	2.479	2.565	2.491	2.362	2.362		
	Ebn at Noon	802	846	927	906	901	884	874	864	857	850	798	766	766		
	Edh at Noon	70	99	96	110	113	116	114	106	90	69	58	57	57		
(44) All-Sky Solar Radiation	RadAvg	1.37	2.57	3.77	4.79	5.39	5.67	6.04	5.03	3.71	2.18	1.21	0.97	0.97		
	RadStd	0.15	0.20	0.37	0.44	0.36	0.45	0.30	0.37	0.30	0.31	0.15	0.11	0.11		

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page